

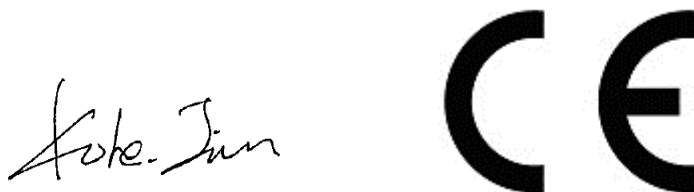
TEST REPORT

Application No.: GZCR2104020061HS
Applicant: Guangdong Invitop Technology Co., Ltd.
Address of Applicant: 2F-201 & 3F, Area A1, Minsen Information Technology Industrial Park, No.8, East of Jinsan Avenue, Sanjiao Town, Zhongshan City, Guangdong Province
Manufacturer: The same as Applicant
Address of Manufacturer: The same as Applicant
Factory: The same as Applicant
Address of Factory: The same as Applicant
Equipment Under Test (EUT):
EUT Name: Dehumidifier
Model No.: TAURUS-RCS20(Y)-N7-1, LEO-RCS20(Y)-N7-1, LIBRA-RCS20(Y)-N7-1, TAURUS-RCS25(Y)-N7-1, LEO-RCS25(Y)-N7-1, LIBRA-RCS25(Y)-N7-1 ♣
 ♣ Please refer to section 2 of this report which indicates which model was actually tested and which were electrically identical.
Standard(s) : EN 301 489-1 V2.2.3
 EN 301 489-17 V3.2.4
Date of Receipt: 2021-04-06
Date of Test: 2021-04-14 to 2021-06-19
Date of Issue: 2021-07-15

Test Result:	Pass*
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* In the configuration tested, the EUT complied with the standards specified above.

The CE mark as shown below can be used, under the responsibility of the manufacturer, after completion of an EU Declaration of Conformity and compliance with all relevant EU Directives.



Kobe Jian
EMC Laboratory Manager



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Revision Record				
Version	Chapter	Date	Modifier	Remark
01		2021-07-15		Original

Authorized for issue by:			
Tested By			
		Lily Kuang/Project Engineer	
Reviewed By			
		Jerry Chan/Reviewer	

2 Test Summary

Emission Part				
Item	Standard	Method	Requirement	Result
Conducted Emissions at AC Mains Power Port (150kHz-30MHz)	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 55032:2015	Class B	Pass
Voltage Fluctuations and Flicker	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-3-3:2013	Clause 5 of EN 61000-3-3	Pass
Harmonic Current Emission	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-3-2:2014	Class A	Pass

Immunity Part				
Item	Standard	Method	Requirement	Result
Radiated Immunity (80MHz-6GHz)	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-4-3:2006 +A1:2008+A2:2010	3V/m, 80%, 1kHz Amp. Mod.	Pass
Surge at AC Mains Power Port	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-4-5:2014 +A1:2017	1.2/50μs Tr/Td; 1kV Line to Line	Pass
Conducted Immunity at AC Mains Power Port (150kHz-80MHz)	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-4-6:2014	3Vrms (emf), 80%, 1kHz Amp. Mod.	Pass
Voltage Dips and Interruptions	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-4-11:2004 +A1:2017	0 % UT for 0.5per; 0 % UT for 1per; 0 % UT for 250per; 70 % UT for 25per; UT is Supply Voltage	Pass
Electrostatic Discharge	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-4-2:2009	4kV Contact Discharge 8kV Air Discharge	Pass
Electrical Fast Transients Burst at AC Mains Power Port	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4	EN 61000-4-4:2012	1kV; 5/50ns Tr/Td; 5kHz Repetition Frequency	Pass

Note:

E.U.T./EUT means Equipment Under Test.



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SGS-CSTC Standards Technical Services Co., Ltd. No.198 Kazhu Road, Sciencetech Park, Guangzhou Economic & Technology Development District, Guangzhou, China 510663 t (86-20) 82155555 f (86-20) 82075058 www.sgsgroup.com.cn
Guangzhou Branch, EEC Laboratory. 中国·广州·经济技术开发区科学城科珠路198号 邮编: 510663 t (86-20) 82155555 f (86-20) 82075058 sgs.china@sgs.com

Pass means the test result passed the test standard requirement, please find the detailed decision rule in the report relative section.

♣ **Model No.:** TAURUS-RCS20(Y)-N7-1, LEO-RCS20(Y)-N7-1, LIBRA-RCS20(Y)-N7-1,
TAURUS-RCS25(Y)-N7-1, LEO-RCS25(Y)-N7-1, LIBRA-RCS25(Y)-N7-1

According to the declaration from the applicant, the electrical circuit design, layout, components used and internal wiring were identical for all models, with only difference as below table shown.

Model	Compressor, capacity and rated power	Model	Control Board
TAURUS-RCS20(Y)-N7-1	The same 270W	TAURUS-RCS20(Y)-N7-1	The same
LEO-RCS20(Y)-N7-1		TAURUS-RCS25(Y)-N7-1	
LIBRA-RCS20(Y)-N7-1		LEO-RCS20(Y)-N7-1	The same
TAURUS-RCS25(Y)-N7-1	The same 320W	LEO-RCS25(Y)-N7-1	
LEO-RCS25(Y)-N7-1		LIBRA-RCS20(Y)-N7-1	The same
LIBRA-RCS25(Y)-N7-1		LIBRA-RCS25(Y)-N7-1	

Models LEO-RCS20(Y)-N7-1, LIBRA-RCS20(Y)-N7-1 and TAURUS-RCS25(Y)-N7-1 were tested in this report.

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4 General Information

4.1 Details of E.U.T.

Power supply:	AC 220-240V,50-60Hz
Cable(s):	AC mains (unshielded, 1.6 m)
Operation Frequency:	802.11b/g/n(HT20): 2412MHz to 2472MHz, 802.11n(HT40): 2422MHz to 2462MHz
Modulation Type:	802.11b: DSSS (CCK, DQPSK, DBPSK), 802.11g/n: OFDM (64QAM, 16QAM, QPSK, BPSK)
Channel Spacing:	5MHz
Number of Channels:	802.11b/g/n(HT20): 13,

4.2 Description of Support Units

Description	Manufacturer	Model No.	Serial No.
Mobile Phone	SANSUNG	SM-9508	R28K110W9JV

4.3 Measurement Uncertainty

Test Item	Measurement Uncertainty
Conducted Emissions at AC Mains Power Port (150kHz-30MHz)	3.12dB
Radiated Emissions (30MHz-1GHz)	5.06dB (3m) 4.46dB (10m)
Remark: The U_{lab} (lab Uncertainty) is less than U_{cispr} (CISPR Uncertainty), so the test results – compliance is deemed to occur if no measured disturbance level exceeds the disturbance limit; – non-compliance is deemed to occur if any measured disturbance level exceeds the disturbance limit.	

4.4 Test Location

All tests were performed at:

SGS-CSTC Standards Technical Services Co., Ltd., Guangzhou Branch EMC Laboratory,
198 Kezhu Road, Sciencetech Park, Guangzhou Economic & Technology Development District,
Guangzhou, China 510663

Tel: +86 20 82155555

Fax: +86 20 82075059

No tests were sub-contracted.



4.5 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

● **NVLAP (Lab Code: 200611-0)**

SGS-CSTC Standards Technical Services Co., Ltd., Guangzhou EMC Laboratory is accredited by the National Voluntary Laboratory Accreditation Program (NVLAP/NIST). NVLAP Code: 200611-0.

The report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the Federal Government.

● **ACMA**

SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory can also perform testing for the Australian/New Zealand Regulatory Compliance Mark (RCM).

● **SGS UK(Certificate No.: 32), SGS-TUV SAARLAND and SGS-FIMKO**

Have approved SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory as a supplier of EMC TESTING SERVICES and SAFETY TESTING SERVICES.

● **CNAS (Lab Code: L0167)**

SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory has been assessed and in compliance with CNAS-CL01:2018 accreditation criteria for testing laboratories (identical to ISO/IEC 17025:2017 General Requirements) for the Competence of Testing Laboratories.

● **FCC Recognized Accredited Test Firm(Registration No.: 486818)**

SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory has been accredited and fully described in a report filed with the (FCC) Federal Communications Commission. The acceptance letter from the FCC is maintained in our files. Designation Number: CN5016, Test Firm Registration Number: 486818.

● **ISED (Registration No.: 4620B, CAB identifier: CN0052)**

SGS-CSTC Standards Technical Services Co., Ltd., has been registered by Innovation Science and Economic Development Canada for Wireless Device Testing laboratories to test to Canadian radio equipment requirements. Registration No. 4620B, CAB identifier: CN0052.

● **VCCI (Registration No.: R-12460, C-12584, G-20107 and T-11179)**

The 10m Semi-anechoic chamber, 966 Anechoic Chamber and Shielded Room of SGS-CSTC Standards Technical Services Co., Ltd. have been registered in accordance with the Regulations for Voluntary Control Measures with Registration No.: R-12460, C-12584, G-20107 and T-11179 respectively.

● **CBTL (Lab Code: TL129)**

SGS-CSTC Standards Technical Services Co., Ltd., E&E Laboratory has been assessed and fully comply with the requirements of ISO/IEC 17025:2017, the Basic Rules, IECEE 01 and Rules of procedure IECEE 02, and the relevant IECEE CB-Scheme Operational documents.



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4.6 Deviation from Standards

None

4.7 Abnormalities from Standard Conditions

None

4.8 EMS Monitor

Visual: Monitored the LED lighting of the EUT

Audio: N/A

Other: Monitored the Spectrum Analyser for any unintentional responses



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5 Equipment List

Conducted Emissions at AC Mains Power Port (150kHz-30MHz)					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
EMI Test Receiver (ESCS30)	Rohde & Schwarz	ESCS30	EMC0506	2020-11-13	2021-11-12
Shielding Room	ChangZhou ZhongYu	8m x 3m x 3.8m	EMC0306	N/A	N/A
Two-Line V-Network ENV216 (EMC0118)	Rohde & Schwarz	ENV216	EMC0118	2021-01-08	2022-01-06
LISN ENV216 (EMC2135)	Rohde & Schwarz	ENV216	EMC2135	2020-09-25	2021-09-24
Coaxial Cable	HangTianXing	2m	EMC0107	2020-09-09	2022-09-08
Test Software E3c	Audix	Ver. 5.4.1221b	GZE100-62	N/A	N/A

Radiated Emissions (30MHz-1GHz)					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
EMI Test Receiver ESIB26	Rohde & Schwarz	ESIB26	EMC0522	2021-01-08	2022-01-07
Chamber cable	HangTianXing	N/A	EMC0542	2019-06-28	2021-06-27
Trilog Broadband Antenna 30MHz-1GHz (SEM003-18)	SCHWARZBECK MESS-ELEKTRONIK	VULB 9168	SEM003-18	2019-02-22	2022-02-22
Amplifier	HP	8447F	EMC2065	2020-05-26	2021-05-25
High Pass Filter (915MHz)	FSY MICROWAVE	HM1465-9SS	EMC2079	2021-01-08	2022-01-07
10m Semi-Anechoic Chamber	ETS	N/A	EMC0530	2019-10-20	2022-10-19
Test Software E3	Audix	Ver.6.120110a	GZE100-61	N/A	N/A

Voltage Fluctuations and Flicker					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
AC Power Source	California	50001iX	EMC0608	2021-03-27	2022-03-26
Power Analyzer	California	PACS	EMC0607	2021-03-27	2022-03-26
Test Software CTS4	California	Ver 4.14.0	GZE100-66	N/A	N/A



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Radiated Immunity (80MHz-6GHz)					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
Compact 3m Semi-Anechoic Chamber	ChangZhou ZhongYu	N/A	EMC0525	2019-10-20	2022-10-19
Laser Probe Interface	RF Microwave Instrumentation	FI7000	EMC2089	N/A	N/A
Open Switch And Control Unit	Rohde & Schwarz	OSP130	EMC2090	N/A	N/A
Broadband Amplifier (80MHz~1GHz/250W)	Rohde & Schwarz	BBA150	EMC2091	2021-01-09	2022-01-08
Broadband Amplifier (800MHz~3GHz/110W)	Rohde & Schwarz	BBA150	EMC2092	2021-03-02	2022-03-01
Signal Generator	Rohde & Schwarz	SMB100A	EMC2093	2021-01-09	2022-01-08
Laser Probe	RF Microwave Instrumentation	FL7006	EMC2094	2021-03-02	2022-03-01
NRP-Z91 Power Sensor 6GHz	Rohde & Schwarz	NPR-Z91	EMC2095	2021-01-09	2022-01-08
NRP-Z91 Power Sensor 6GHz	Rohde & Schwarz	NPR-Z91	EMC2096	2021-01-09	2022-01-08
High-Gain Log-preiodic Antenna	Rohde & Schwarz	HL046E	EMC2097	2019-02-15	2022-02-14
Broadband Amplifier (2.5~6GHz/30W)	Rohde & Schwarz	BBA150	EMC2105	2020-09-18	2021-09-17
RI Cable	Rohde & Schwarz	7m	EMC2098	2020-05-24	2021-05-23
Oscilloscope	Tektronix	TDS3052C	EMC2055	2020-12-10	2021-12-09
Audio Analyzer	Keysight	U8903B	EMC2180	2020-09-18	2021-09-17
Monitor System	Mitsubish Corp.	M-0552AB	EMC0909	N/A	N/A
Test Software EMC32	Rohde & Schwarz	Ver. 9.26.00	GZE100-63	N/A	N/A

Surge at AC Mains Power Port					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
EMC Immunity Test System	TESEQ AG	NSG 3060 CDN3061 INA 6502 CIB CND3425	EMC2072	2021-01-08	2022-01-07
Oscilloscope	Tektronix	TDS3052C	EMC2055	2020-12-10	2021-12-09
Test Software WIN 3000	TESEQ AG	Ver 1.3.2	GZE100-68	N/A	N/A

Conducted Immunity at AC Mains Power Port (150kHz-80MHz)					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
Test System for Conducted and Radiated Immunity	TESEQ AG	NSG 4070B-80	EMC2115	2020-11-30	2021-11-29
Test Software NSG4070_Ctrl1	TESEQ AG	Ver.1.3.0.1	GZE100-72	N/A	N/A
Dual Directional coupler	Werlatone Inc.	C1795	EMC1105	2020-05-26	2021-05-25
Oscilloscope	Tektronix	TDS3052C	EMC2055	2020-12-10	2021-12-09
CDN M2/M3	Elektronik-Feinmechanik	L-801:M2/M3	EMC2048	2020-08-21	2022-08-20
CDN M2	Schaffner Chase	CDN-M2-16	EMC1107	2020-10-23	2023-10-22
Audio Analyzer	Keysight	U8903B	EMC2180	2020-09-18	2021-09-17

Voltage Dips and Interruptions					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
EMC Immunity Test System	TESEQ AG	NSG 3060 CDN3061 INA 6502 CIB CND3425	EMC2072	2021-01-08	2022-01-07
Oscilloscope	Tektronix	TDS3052C	EMC2055	2020-12-10	2021-12-09
Test Software WIN 3000	TESEQ AG	Ver 1.3.2	GZE100-68	N/A	N/A

Electrostatic Discharge					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
ESD Simulator (TESEQ AG)	TESEQ AG	NSG 435	EMC2071	2020-06-30	2021-06-30
ESD Simulator (EMTEST)	EMTEST	NX30	EMC2186	2021-02-27	2022-02-26
ESD Ground Plane	SGS-EMC	3m x 3m	EMC0804	N/A	N/A
Aneroid Barometer	Shanghai Meteorological Instrument Factory Co., Ltd.	YM3	EMC2181	2020-12-03	2021-12-02
Temperature & Humidity	Shanghai Meteorological Instrument Factory Co., Ltd.	ZJ1-2B	EMC0078	2020-07-04	2021-07-03

Electrical Fast Transients Burst at AC Mains Power Port					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
EMC Immunity Test System	TESEQ AG	NSG 3060 CDN3061 INA 6502 CIB CND3425	EMC2072	2021-01-08	2022-01-07
Oscilloscope	Tektronix	TDS3052C	EMC2055	2020-12-10	2021-12-09
Test Software WIN 3000	TESEQ AG	Ver 1.3.2	GZE100-68	N/A	N/A

General used equipment					
Equipment	Manufacturer	Model No	Inventory No	Cal Date	Cal Due Date
DMM	Fluke	73	EMC0006	2020-07-09	2021-07-08
DMM	Fluke	73	EMC0007	2020-07-09	2021-07-08



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6 Emission Test Results

6.1 Conducted Emissions at AC Mains Power Port (150kHz-30MHz)

Test Requirement: EN 301 489-1 V2.2.3

Test Method: EN 55032:2015

Limit:

Frequency of emission(MHz)	Conducted limit(dBμV)	
	Quasi-peak	Average
0.15-0.5	66 to 56*	56 to 46*
0.5-5	56	46
5-30	60	50

*Decreases with the logarithm of the frequency.

Detector: Peak for pre-scan (9kHz resolution bandwidth) 0.15M to 30MHz

6.1.1 E.U.T. Operation

Operating Environment:

Temperature: 25.8 °C

Humidity: 62.1 % RH

Atmospheric Pressure: 1010 mbar

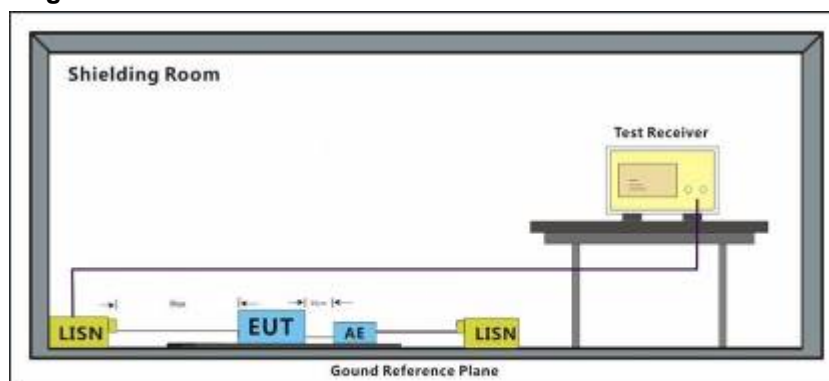
6.1.2 Test Mode Description

Pre-scan / Mode Description

Final test Code

Final test 02 Operation(WiFi)_Pair the device with Smartphone

6.1.3 Test Setup Diagram



6.1.4 Measurement Procedure and Data

Frequency range: 150KHz-30MHz

An initial pre-scan was performed with peak detector. Quasi-Peak or Average measurement were performed at the frequencies with maximized peak emission were detected.



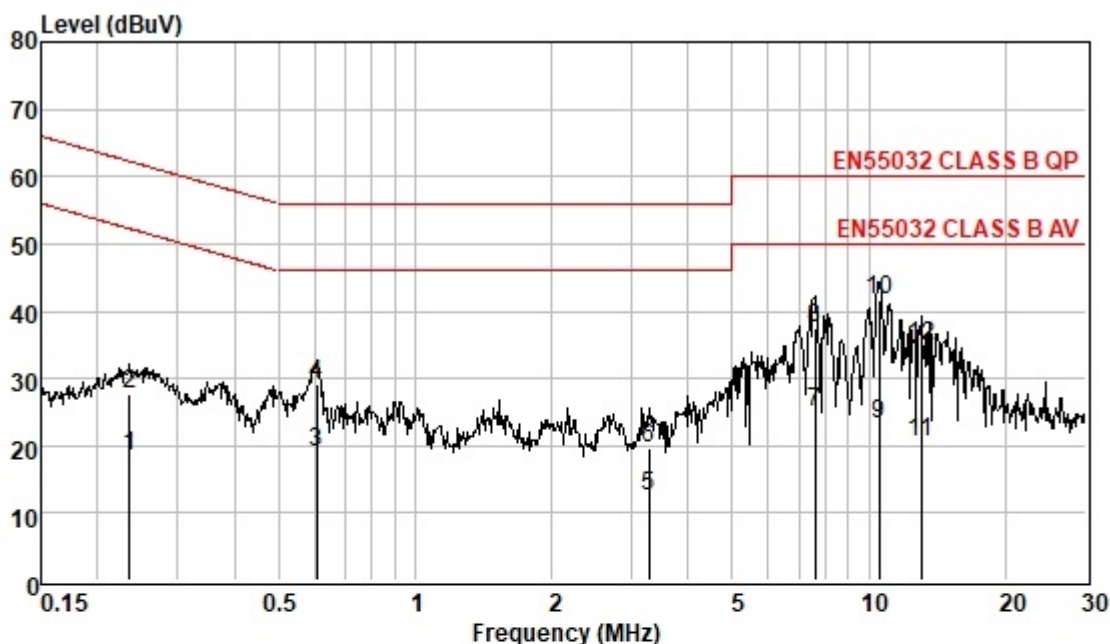
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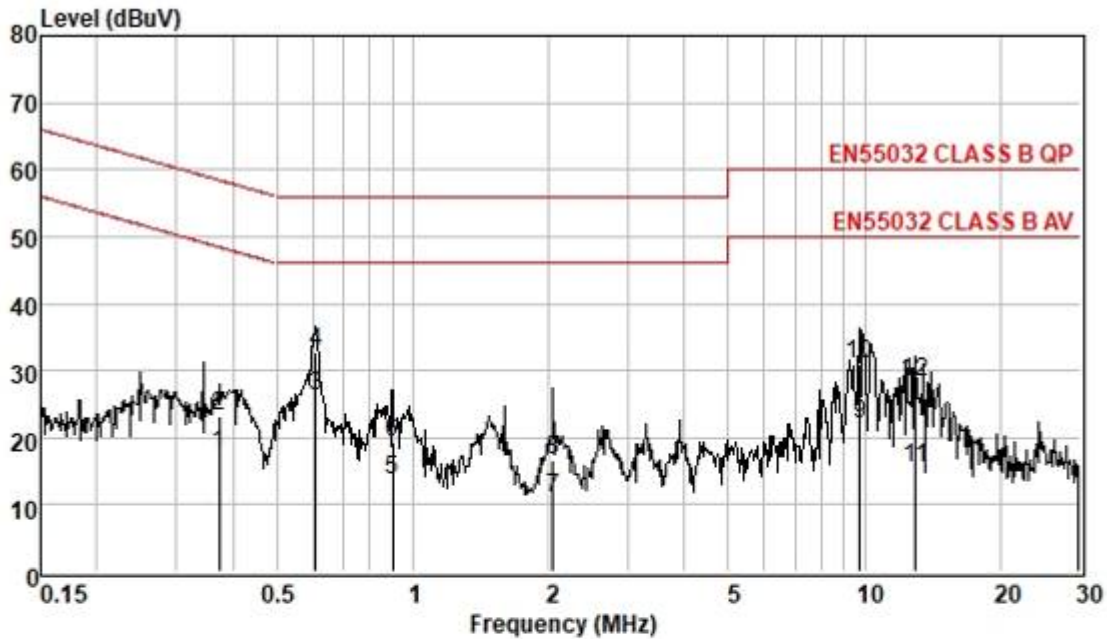
No.198 Kazhu Road, Science Park, Guangzhou Economic & Technology Development District, Guangzhou, China 510663 t (86-20) 82155555 f (86-20) 82075058 www.sgsgroup.com.cn
 中国·广州·经济技术开发区科学城科珠路198号 邮编: 510663 t (86-20) 82155555 f (86-20) 82075058 sgs.china@sgs.com

Test Mode: 02; Line: Live line; Model: LEO-RCS20(Y)-N7-1

Pol : LINE
Mode : M1
Model :

Freque	Read	Cable	LISN	Measured	Limit	Over	Remark
nc	Level	Loss	Factor	Level	Line	Limit	
MHz	dBuV	dB	dB	dBuV	dBuV	dB	
0.24	8.83	0.06	9.62	18.51	52.26	-33.75	Average
0.24	17.87	0.06	9.62	27.55	62.26	-34.71	QP
0.61	9.32	0.07	9.63	19.02	46.00	-26.98	Average
0.61	19.53	0.07	9.63	29.23	56.00	-26.77	QP
3.28	2.76	0.15	9.62	12.53	46.00	-33.47	Average
3.28	9.95	0.15	9.62	19.72	56.00	-36.28	QP
7.61	15.01	0.21	9.67	24.89	50.00	-25.11	Average
7.61	27.46	0.21	9.67	37.34	60.00	-22.66	QP
10.51	13.36	0.24	9.70	23.30	50.00	-26.70	Average
10.51	31.81	0.24	9.70	41.75	60.00	-18.25	QP
12.99	10.61	0.28	9.72	20.61	50.00	-29.39	Average
12.99	24.91	0.28	9.72	34.91	60.00	-25.09	QP

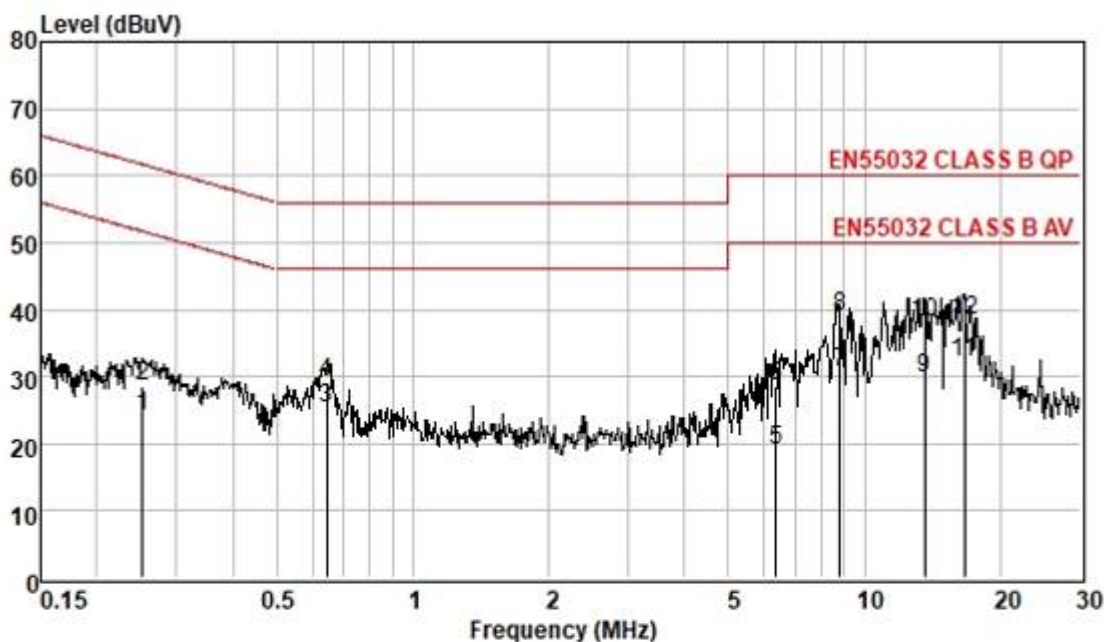
Test Mode: 02; Line: Live line; Model: LIBRA-RCS20(Y)-N7-1



Pol : LINE
Mode : RCS20(Y)N7-1
Model :

Frequency MHz	Read Level dBuV	Cable Loss dB	LISN Factor dB	Measured Level dBuV	Limit Line dBuV	Over Limit dB	Remark
0.37	7.82	0.06	9.63	17.51	48.43	-30.92	Average
0.37	13.40	0.06	9.63	23.09	58.43	-35.34	QP
0.61	16.36	0.07	9.63	26.06	46.00	-19.94	Average
0.61	22.97	0.07	9.63	32.67	56.00	-23.33	QP
0.90	4.03	0.07	9.62	13.72	46.00	-32.28	Average
0.90	9.62	0.07	9.62	19.31	56.00	-36.69	QP
2.04	1.21	0.12	9.62	10.95	46.00	-35.05	Average
2.04	6.85	0.12	9.62	16.59	56.00	-39.41	QP
9.76	12.00	0.23	9.69	21.92	50.00	-28.08	Average
9.76	21.05	0.23	9.69	30.97	60.00	-29.03	QP
12.92	5.49	0.28	9.72	15.49	50.00	-34.51	Average
12.92	18.30	0.28	9.72	28.30	60.00	-31.70	QP

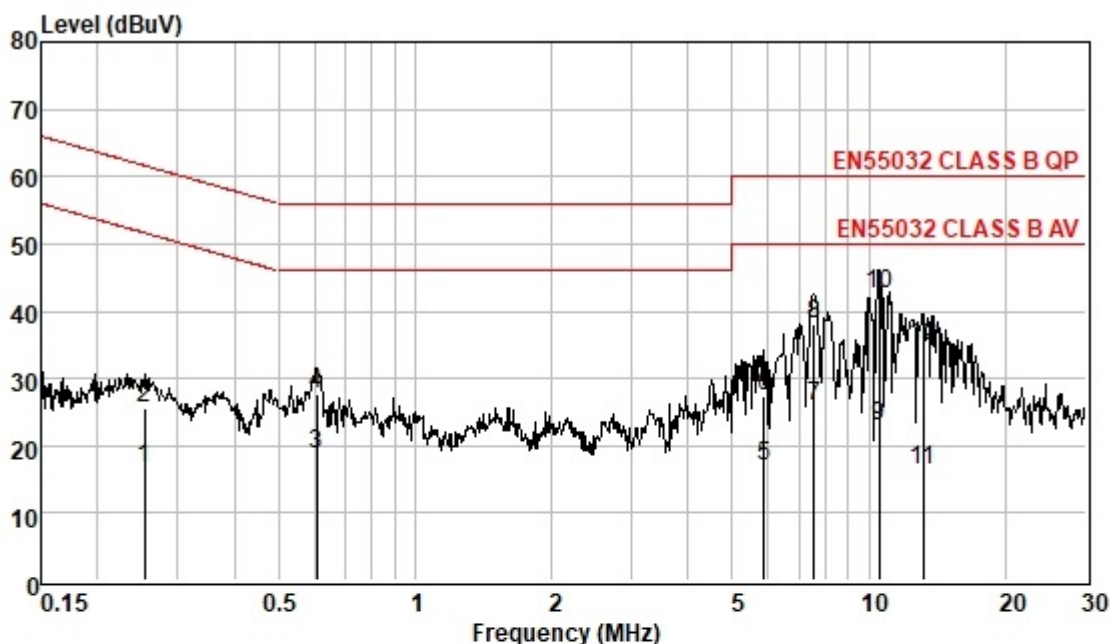
Test Mode: 02; Line: Live line; Model: TAURUS-RCS25(Y)-N7-1



Pol : LINE
Mode : M2
Model :

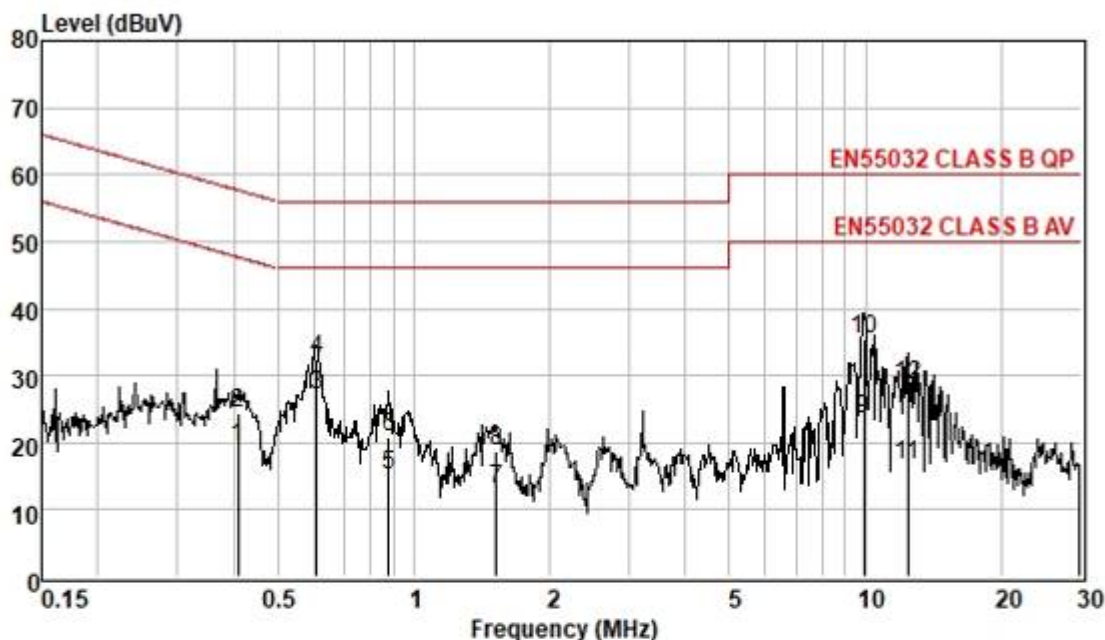
Frequency MHz	Read Level dBUV	Cable Loss dB	LISN Factor dB	Measured Level dBUV	Limit Line dBUV	Over Limit dB	Remark
0.25	14.54	0.06	9.62	24.22	51.69	-27.47	Average
0.25	18.75	0.06	9.62	28.43	61.69	-33.26	QP
0.64	15.49	0.07	9.63	25.19	46.00	-20.81	Average
0.64	19.51	0.07	9.63	29.21	56.00	-26.79	QP
6.35	9.03	0.20	9.67	18.90	50.00	-31.10	Average
6.35	18.79	0.20	9.67	28.66	60.00	-31.34	QP
8.82	20.72	0.22	9.68	30.62	50.00	-19.38	Average
8.82	29.01	0.22	9.68	38.91	60.00	-21.09	QP
13.55	19.67	0.29	9.73	29.69	50.00	-20.31	Average
13.55	28.05	0.29	9.73	38.07	60.00	-21.93	QP
16.57	21.91	0.33	9.75	31.99	50.00	-18.01	Average
16.57	28.34	0.33	9.75	38.42	60.00	-21.58	QP

Test Mode: 02; Line: Neutral Line; Model: LEO-RCS20(Y)-N7-1

Pol : NEUTRAL
Mode : M1
Model :

Frequency MHz	Read Level dBuV	Cable Loss dB	LISN Factor dB	Measured Level dBuV	Limit Line dBuV	Over Limit dB	Remark
0.25	7.09	0.06	9.55	16.70	51.64	-34.94	Average
0.25	15.89	0.06	9.55	25.50	61.64	-36.14	QP
0.61	9.23	0.07	9.54	18.84	46.00	-27.16	Average
0.61	17.91	0.07	9.54	27.52	56.00	-28.48	QP
5.87	7.22	0.19	9.57	16.98	50.00	-33.02	Average
5.87	17.65	0.19	9.57	27.41	60.00	-32.59	QP
7.57	16.21	0.21	9.58	26.00	50.00	-24.00	Average
7.57	28.21	0.21	9.58	38.00	60.00	-22.00	QP
10.51	12.99	0.24	9.60	22.83	50.00	-27.17	Average
10.51	32.61	0.24	9.60	42.45	60.00	-17.55	QP
13.13	6.53	0.28	9.62	16.43	50.00	-33.57	Average
13.13	25.02	0.28	9.62	34.92	60.00	-25.08	QP

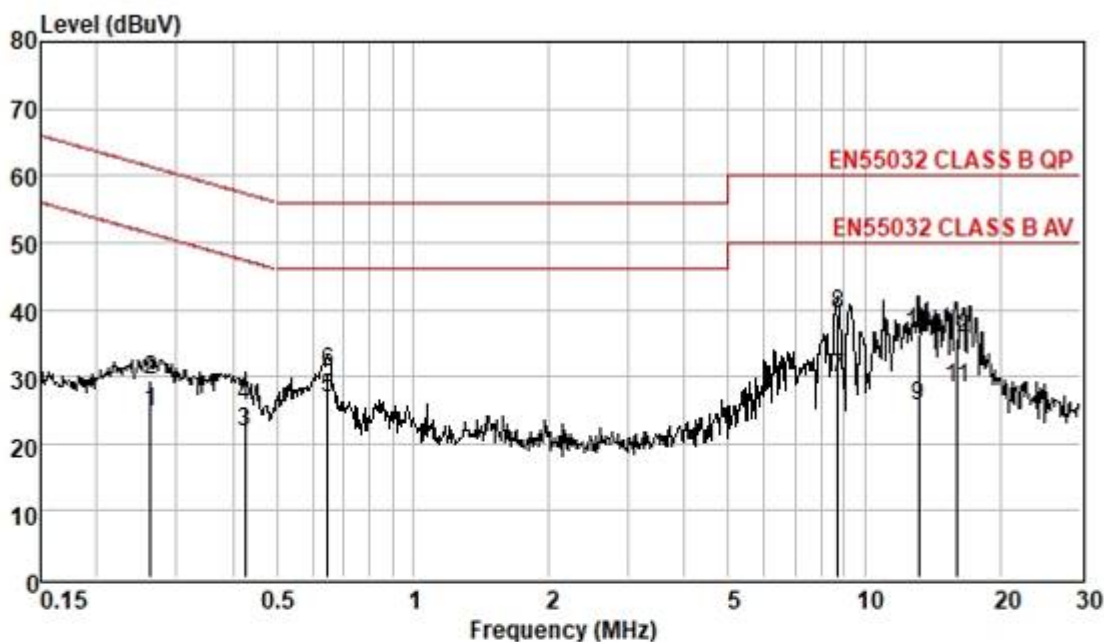
Test Mode: 02; Line: Neutral Line; Model: LIBRA-RCS20(Y)-N7-1



Pol : NEUTRAL
Mode : RCS20(Y)N7-1
Model :

Frequency MHz	Read Level dBuV	Cable Loss dB	LISN Factor dB	Measured Level dBuV	Limit Line dBuV	Over Limit dB	Remark
0.41	9.70	0.06	9.56	19.32	47.68	-28.36	Average
0.41	14.68	0.06	9.56	24.30	57.68	-33.38	QP
0.61	17.49	0.07	9.54	27.10	46.00	-18.90	Average
0.61	22.91	0.07	9.54	32.52	56.00	-23.48	QP
0.88	5.53	0.07	9.55	15.15	46.00	-30.85	Average
0.88	11.24	0.07	9.55	20.86	56.00	-35.14	QP
1.52	3.44	0.10	9.55	13.09	46.00	-32.91	Average
1.52	8.97	0.10	9.55	18.62	56.00	-37.38	QP
9.91	13.71	0.23	9.59	23.53	50.00	-26.47	Average
9.91	25.65	0.23	9.59	35.47	60.00	-24.53	QP
12.38	6.77	0.27	9.62	16.66	50.00	-33.34	Average
12.38	18.61	0.27	9.62	28.50	60.00	-31.50	QP

Test Mode: 02; Line: Neutral Line; Model: TAURUS-RCS25(Y)-N7-1



Pol : NEUTRAL
Mode : M2
Model :

Frequency MHz	Read Level dBUV	Cable Loss dB	LISN Factor dB	Measured Level dBUV	Limit Line dBUV	Over Limit dB	Remark
0.26	14.95	0.06	9.55	24.56	51.34	-26.78	Average
0.26	19.69	0.06	9.55	29.30	61.34	-32.04	QP
0.43	11.99	0.06	9.56	21.61	47.33	-25.72	Average
0.43	15.95	0.06	9.56	25.57	57.33	-31.76	QP
0.65	17.19	0.07	9.54	26.80	46.00	-19.20	Average
0.65	21.09	0.07	9.54	30.70	56.00	-25.30	QP
8.73	19.90	0.22	9.59	29.71	50.00	-20.29	Average
8.73	29.31	0.22	9.59	39.12	60.00	-20.88	QP
13.13	15.68	0.28	9.62	25.58	50.00	-24.42	Average
13.13	26.28	0.28	9.62	36.18	60.00	-23.82	QP
16.05	18.35	0.32	9.65	28.32	50.00	-21.68	Average
16.05	25.59	0.32	9.65	35.56	60.00	-24.44	QP

7 Emission Test Results

7.1 Radiated Emissions (30MHz-1GHz)

N/A, since there are not any ancillary equipment connected to the radio equipment. The Radiated Emission test is only applicable to ancillary equipment not incorporated in the radio equipment and intended to be measured on a stand-alone basis, as declared by the manufacturer



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7.2 Voltage Fluctuations and Flicker

Test Requirement: EN 301 489-1 V2.2.3

Test Method: EN 61000-3-3:2013

7.2.1 E.U.T. Operation

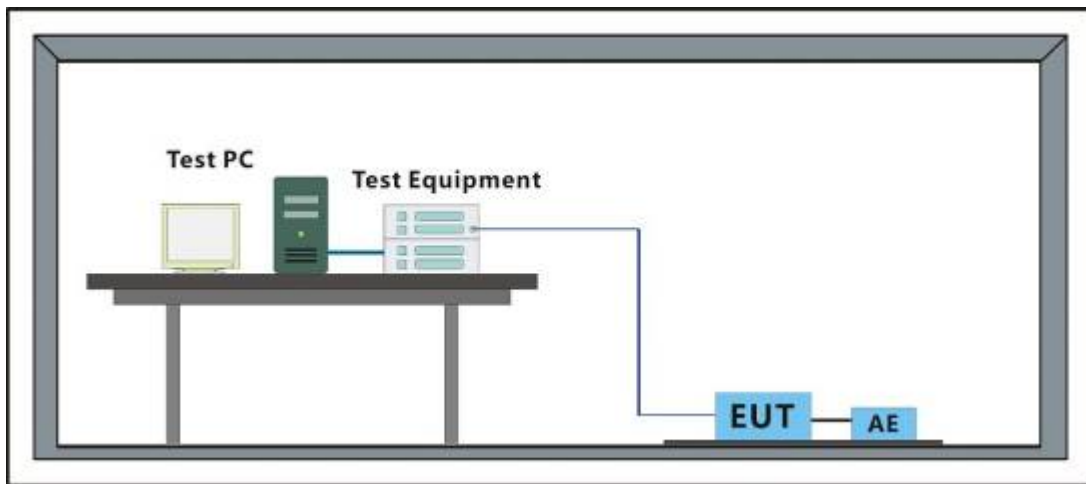
Operating Environment:

Temperature: 21 °C Humidity: 52 % RH Atmospheric Pressure: 1010 mbar

7.2.2 Test Mode Description

Pre-scan /	Mode	Description
Final test	Code	
Final test	02	Operation(WiFi)_Pair the device with Smartphone

7.2.3 Test Setup Diagram



7.2.4 Measurement Procedure and Data

Test Mode: 02; Model: LEO-RCS20(Y)-N7-1

Flicker Test Summary per EN/IEC61000-3-3 (Run time)

Test category: dt,dmax,dc and Pst (European limits)
 Test duration (min): 10

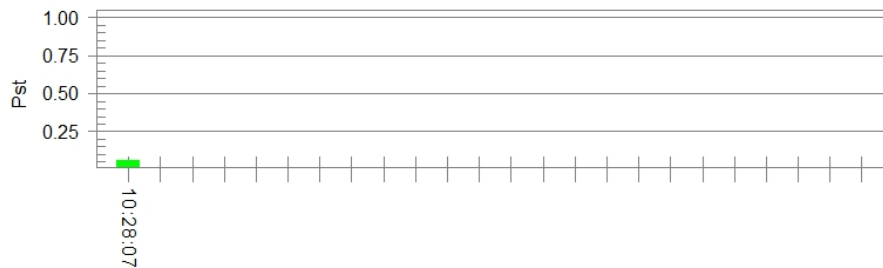
Test Margin: 100

Test Result: Pass

Status: Test Completed

Pst, and limit line

European Limits



Parameter values recorded during the test:

Vrms at the end of test (Volt): 229.95

T-max (mS): 0

Highest dc (%): 0.00

Highest dmax (%): 0.00

Highest Pst (10 min. period): 0.064

Test limit (mS): 500.0 Pass

Test limit (%): 3.30 Pass

Test limit (%): 4.00 Pass

Test limit: 1.000 Pass



Test Mode: 02; Model: LIBRA-RCS20(Y)-N7-1

Flicker Test Summary per EN/IEC61000-3-3 (Run time)

Test category: dt,dmax,dc and Pst (European limits)

Test Margin: 100

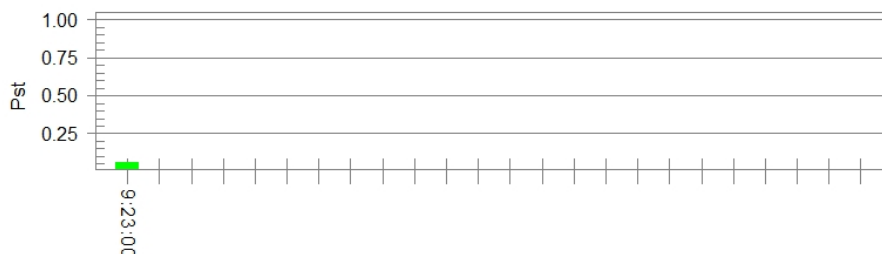
Test duration (min): 10

Test Result: Pass

Status: Test Completed

Pst, and limit line

European Limits



Parameter values recorded during the test:

Vrms at the end of test (Volt): 230.01

T-max (mS): 0

Highest dc (%): 0.00

Highest dmax (%): 0.00

Highest Pst (10 min. period): 0.064

Test limit (mS): 500.0 Pass

Test limit (%): 3.30 Pass

Test limit (%): 4.00 Pass

Test limit: 1.000 Pass



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Test Mode: 02; Model: TAURUS-RCS25(Y)-N7-1

Flicker Test Summary per EN/IEC61000-3-3 (Run time)

Test category: dt,dmax,dc and Pst (European limits)
Test duration (min): 10

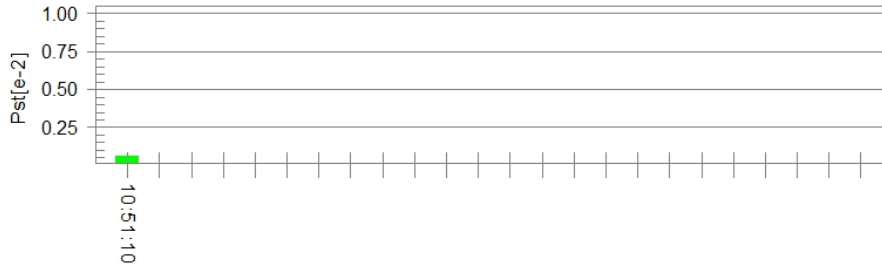
Test Margin: 100

Test Result: Pass

Status: Test Completed

Pst, and limit line

European Limits



Parameter values recorded during the test:

Vrms at the end of test (Volt): 229.99

T-max (mS): 0

Highest dc (%): 0.00

Highest dmax (%): 0.00

Highest Pst (10 min. period): 0.064

Test limit (mS): 500.0 **Pass**

Test limit (%): 3.30 **Pass**

Test limit (%): 4.00 **Pass**

Test limit: 1.000 **Pass**



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7.3 Harmonic Current Emission

Test Requirement: EN 301 489-1 V2.2.3

Test Method: EN 61000-3-2:2014

7.3.1 Conclusion

Frequency range: 100Hz-2KHz

Remark:

Since the EUT was belong to exception of clause 7 and Annex B, according to EN IEC 61000-3-2 figure 1, it was deemed to conform to the requirements of this standard without further testing.

"7 Harmonic current limits

The procedure for applying the limits and assessing the results is shown in Figure 1.

For the following categories of equipment, limits are not specified in this standard:

NOTE 1 Limits may be defined in a future amendment or revision of the standard.

- lighting equipment with a rated power less than but not equal to 5 W;
- equipment with a rated power of 75 W or less, other than lighting equipment;

NOTE 2 This value may be reduced from 75 W to 50 W in the future, subject to approval by National Committees at that time.

- professional equipment with a total rated power greater than 1 kW;
- symmetrically controlled heating elements with a rated power less than or equal to 200 W;
- independent phase control dimmers
 - with a rated power less than or equal to 1 kW when operating incandescent lamps;
 - with a rated power less than or equal to 200 W for trailing edge dimmers, and universal phase control dimmers with the default mode set to trailing edge, when operating lighting equipment other than incandescent lamps;
 - with a rated power less than or equal to 100 W for leading edge dimmers, and universal phase control dimmers without default mode set to trailing edge, when operating lighting equipment other than incandescent lamps.

and

Kitchen machines as listed in the scope of IEC 60335-2-14 are deemed to conform to the harmonic current limits of this standard without further testing.

Please read clause 7 & Annex B of this standard for reference.



8 Immunity Test Results

Performance Criteria Description in EN 301 489-1

Performance criteria for continuous phenomena	During the test, the equipment shall: • continue to operate as intended; • not unintentionally transmit; • not unintentionally change its operating state; • not unintentionally change critical stored data.
Performance criteria for transient phenomena	For all ports and transient phenomena with the exception described below, the following applies: • The application of the transient phenomena shall not result in a change of the mode of operation (e.g. unintended transmission) or the loss of critical stored data. • After application of the transient phenomena, the equipment shall operate as intended. For surges applied to symmetrically operated wired network ports intended to be connected directly to outdoor lines the following criteria applies: • For products with only one symmetrical port intended for connection to outdoor lines, loss of function is allowed, provided the function is self-recoverable, or can be otherwise restored. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost. • For products with more than one symmetrical port intended for connection to outdoor lines, loss of function on the port under test is allowed, provided the function is self-recoverable. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

Performance Criteria Description in EN 301 489-17

Criteria	During Test	After Test
A	Shall operate as intended. (see note). Shall be no loss of function. Shall be no unintentional transmissions.	Shall operate as intended. Shall be no degradation of performance. Shall be no loss of function. Shall be no loss of critical stored data.
B	May be loss of function.	Functions shall be self-recoverable. Shall operate as intended after recovering. Shall be no loss of critical stored data.
C	May be loss of function.	Functions shall be recoverable by the operator. Shall operate as intended after recovering. Shall be no loss of critical stored data.

NOTE: Operate as intended during the test allows a level of degradation in accordance with Minimum performance level.

Minimum performance level

For equipment that supports a PER or FER, the minimum performance level shall be a PER or FER less than or equal to 10 %.

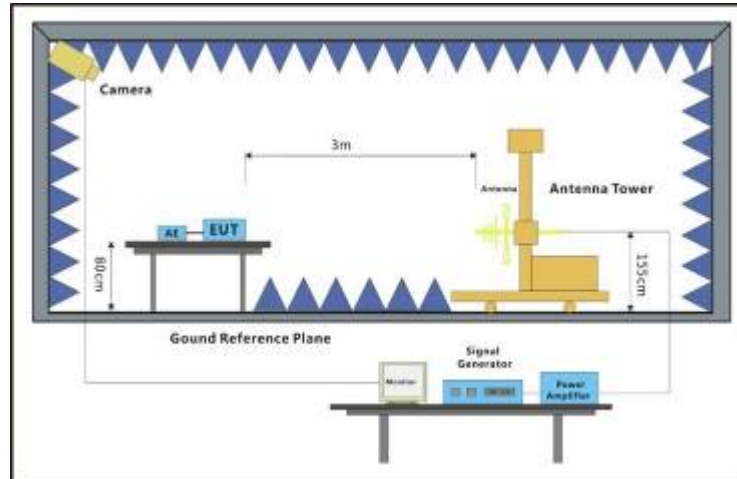
For equipment that does not support a PER or a FER, the minimum performance level shall be no loss of the wireless transmission function needed for the intended use of the equipment.

Performance criteria for Continuous phenomena applied to Transmitters (CT)
The performance criteria A shall apply. Tests shall be repeated with the EUT in standby mode (if applicable) to ensure that unintentional transmission does not occur. In systems using acknowledgement signals, it is recognized that an ACKnowledgement (ACK) or Not ACKnowledgement (NACK) transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.
Performance criteria for Transient phenomena applied to Transmitters (TT)
The performance criteria B shall apply, except for voltage dips of 100 ms and voltage interruptions of 5 000 ms duration, for which performance criteria C shall apply. Tests shall be repeated with the EUT in standby mode (if applicable) to ensure that unintentional transmission does not occur. In systems using acknowledgement signals, it is recognized that an acknowledgement (ACK) or not-acknowledgement (NACK) transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.
Performance criteria for Continuous phenomena applied to Receivers (CR)
The performance criteria A shall apply. Where the EUT is a transceiver, under no circumstances, shall the transmitter operate unintentionally during the test. In systems using acknowledgement signals, it is recognized that an ACK or NACK transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.
Performance criteria for Transient phenomena applied to Receivers (TR)
The performance criteria B shall apply, except for voltage dips of 100 ms and voltage interruptions of 5 000 ms duration for which performance criteria C shall apply. Where the EUT is a transceiver, under no circumstances, shall the transmitter operate unintentionally during the test. In systems using acknowledgement signals, it is recognized that an ACK or NACK transmission may occur, and steps should be taken to ensure that any transmission resulting from the application of the test is correctly interpreted.

8.1 Radiated Immunity (80MHz-6GHz)

Test Requirement: EN 301 489-1 V2.2.3
 Test Method: EN 61000-4-3:2006 +A1:2008+A2:2010

8.1.1 Test Setup Diagram



8.1.2 E.U.T. Operation

Operating Environment:
 Temperature: 23.2 °C Humidity: 52 % RH Atmospheric Pressure: 1010 mbar

8.1.3 Test Mode Description

Pre-scan / Mode	Description
Final test Code	
Final test 02	Operation(WiFi)_Pair the device with Smartphone
Final test 03	Idle:Keep the EUT standby.

8.1.4 Test Condition and Results:

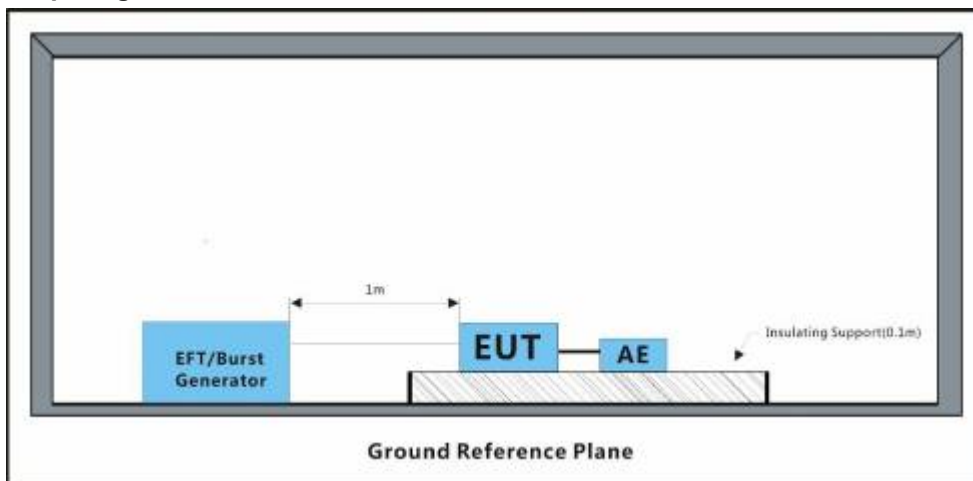
Performance Criterion: A
 Frequency Range: 80MHz to 6GHz
 Antenna Polarisation: Vertical and Horizontal
 Modulation: 1kHz,80% Amp. Mod,1% increment

Frequency	Level (V/m)	EUT Face	Dwell time	Result / Observations
80MHz-6GHz	3	Front	2s	A
80MHz-6GHz	3	Back	2s	A
80MHz-6GHz	3	Left	2s	A
80MHz-6GHz	3	Right	2s	A
80MHz-6GHz	3	Top	2s	A
80MHz-6GHz	3	Underside	2s	A
A: No degradation in the performance of the EUT was observed				

8.2 Surge at AC Mains Power Port

Test Requirement: EN 301 489-1 V2.2.3
 Test Method: EN 61000-4-5:2014 +A1:2017

8.2.1 Test Setup Diagram



8.2.2 E.U.T. Operation

Operating Environment:
 Temperature: 24.4 °C Humidity: 60.6 % RH Atmospheric Pressure: 1010 mbar

8.2.3 Test Mode Description

Pre-scan / Mode	Description
Final test Code	
Final test 02	Operation(WiFi)_Pair the device with Smartphone
Final test 03	Idle:Keep the EUT standby.

8.2.4 Test Condition and Results:

Performance Criterion: B
 Interval: 60s between each surge
 No. of surges: 5 positive, 5 negative at 0°, 90°, 180°, 270°

Test Line	Level (kV)	Polarity	Phase (deg)	Result / Observations
L-N	1	+	0°	A
L-N	1	-	0°	A
L-N	1	+	90°	A
L-N	1	-	90°	A
L-N	1	+	180°	A
L-N	1	-	180°	A
L-N	1	+	270°	A
L-N	1	-	270°	A

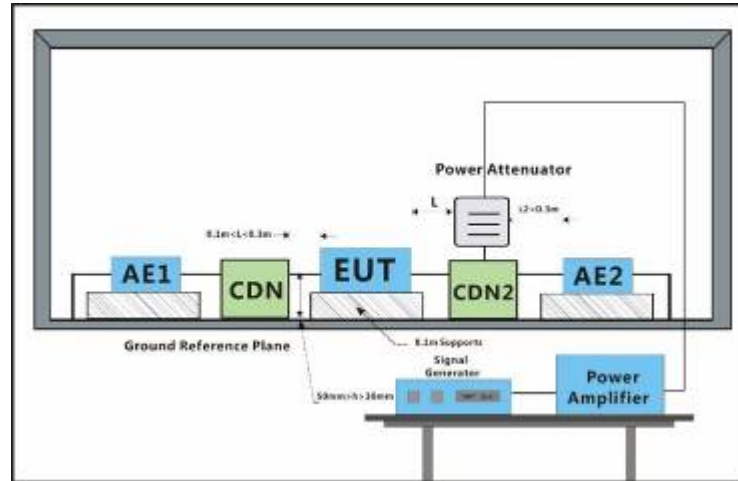
A: No degradation in the performance of the EUT was observed

8.3 Conducted Immunity at AC Mains Power Port (150kHz-80MHz)

Test Requirement: EN 301 489-1 V2.2.3

Test Method: EN 61000-4-6:2014

8.3.1 Test Setup Diagram



8.3.2 E.U.T. Operation

Operating Environment:

Temperature: 24.4 °C

Humidity: 60.7 % RH

Atmospheric Pressure: 1010 mbar

8.3.3 Test Mode Description

Pre-scan / Mode	Description
Final test Code	
Final test 02	Operation(WiFi)_Pair the device with Smartphone
Final test 03	Idle:Keep the EUT standby.

8.3.4 Test Condition and Results:

Performance Criterion: A

Frequency Range: 0.15MHz to 80MHz

Modulation: 80%, 1kHz Amplitude Modulation

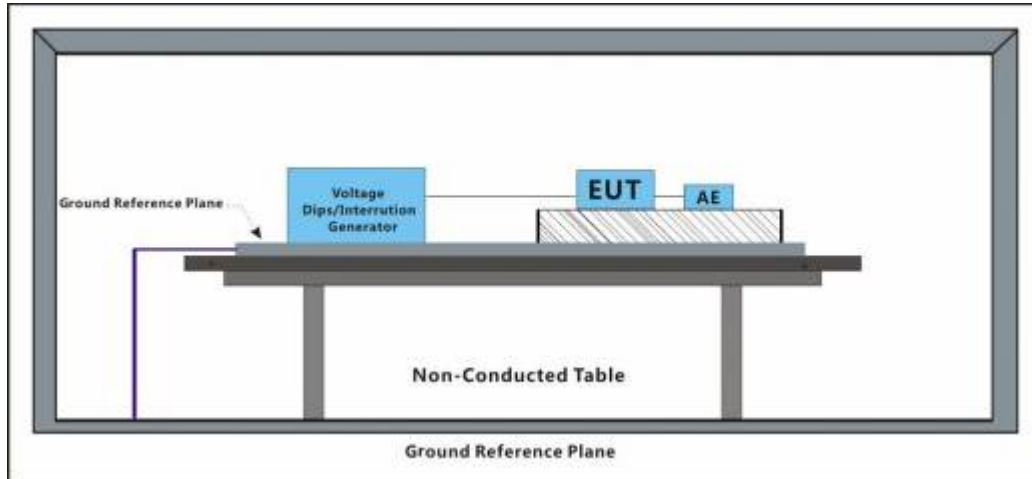
Step Size: 1%

Cable port	Level (Vrms)	CDN/Clamp	Dwell time	Result / Observations
AC power port	3	CDN	2s	A
A: No degradation in the performance of the EUT was observed				

8.4 Voltage Dips and Interruptions

Test Requirement: EN 301 489-1 V2.2.3
 Test Method: EN 61000-4-11:2004 +A1:2017

8.4.1 Test Setup Diagram



8.4.2 E.U.T. Operation

Operating Environment:
 Temperature: 24.4 °C Humidity: 60.7 % RH Atmospheric Pressure: 1010 mbar

8.4.3 Test Mode Description

Pre-scan / Mode	Description
Final test Code	
Final test 02	Operation(WiFi)_Pair the device with Smartphone
Final test 03	Idle:Keep the EUT standby.

8.4.4 Test Condition and Results:

Performance Criterion: 0% of UT (Supply Voltage) for 0.5 Period:B;
 0% of UT for 1 Period:B; 0% of UT for 250 Periods:C;
 70 % of UT for 25 Periods:C
 No. of Dips / Interruptions: 3 per Level
 Time between dropout: 10s

Level % UT	Phase (deg)	Duration	No. of Dips / Interruptions	Result / Observations
0	0°	0.5 Cycle	3	A
0	180°	0.5 Cycle	3	A
0	0°	1 Cycle	3	A
0	180°	1 Cycle	3	A
0	0°	250 Cycles	3	C
0	180°	250 Cycles	3	C
70	0°	25 Cycles	3	A
70	180°	25 Cycles	3	A

A: No degradation in the performance of the EUT was observed

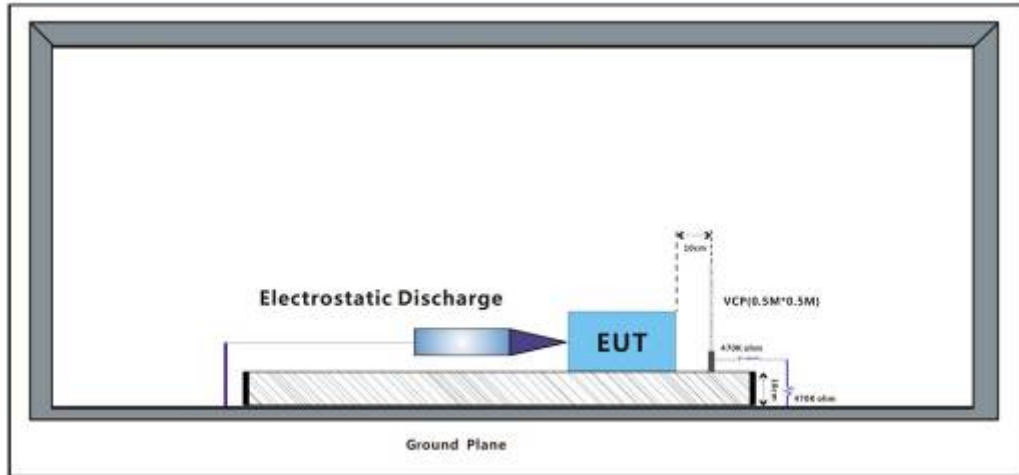
C: During test, the EUT stopped working. After test, it could be recovered manually.

8.5 Electrostatic Discharge

Test Requirement: EN 301 489-1 V2.2.3

Test Method: EN 61000-4-2:2009

8.5.1 Test Setup Diagram



8.5.2 E.U.T. Operation

Operating Environment:

Temperature: 21 °C Humidity: 52 % RH Atmospheric Pressure: 1010 mbar

8.5.3 Test Mode Description

Pre-scan /	Mode	Description
Final test	Code	
Final test	02	Operation(WiFi)_Pair the device with Smartphone
Final test	03	Idle:Keep the EUT standby.

8.5.4 Test Condition and Results:

Performance Criterion: B

Discharge Impedance: 330 Ω /150pF

Number of Discharge: Minimum 10 times at each test point

Discharge Mode: Single Discharge

Discharge Period: 1 second minimum

Test Point 1: All insulated enclosure & seams.

Test Point 2: All accessible metal parts of the enclosure.

Test Point 3: All sides.

Discharge type	Level (kV)	Polarity	Test Point	Result / Observations
Air Discharge	2,4,8	+	1	A
Air Discharge	2,4,8	-	1	A
Contact Discharge	4	+	2	N/A
Contact Discharge	4	-	2	N/A
Horizontal Coupling	4	+	3	N/A
Horizontal Coupling	4	-	3	N/A
Vertical Coupling	4	+	3	A
Vertical Coupling	4	-	3	A

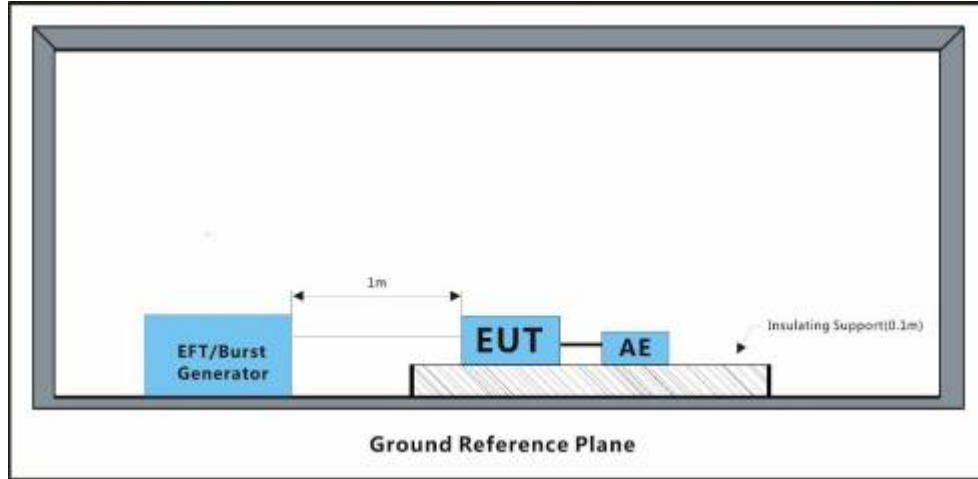
A: No degradation in the performance of the EUT was observed

8.6 Electrical Fast Transients Burst at AC Mains Power Port

Test Requirement: EN 301 489-1 V2.2.3

Test Method: EN 61000-4-4:2012

8.6.1 Test Setup Diagram



8.6.2 E.U.T. Operation

Operating Environment:

Temperature: 24.4 °C

Humidity: 60.7 % RH

Atmospheric Pressure: 1010 mbar

8.6.3 Test Mode Description

Pre-scan / Mode	Description
Final test Code	
Final test 02	Operation(WiFi)_Pair the device with Smartphone
Final test 03	Idle:Keep the EUT standby.

8.6.4 Test Condition and Results:

Performance Criterion: B

Repetition Frequency: 5kHz

Burst Period: 300ms

Test Line	Level (kV)	Polarity	CDN/Clamp	Result / Observations
AC mains power port	1	+	CDN	A
AC mains power port	1	-	CDN	A
A: No degradation in the performance of the EUT was observed				

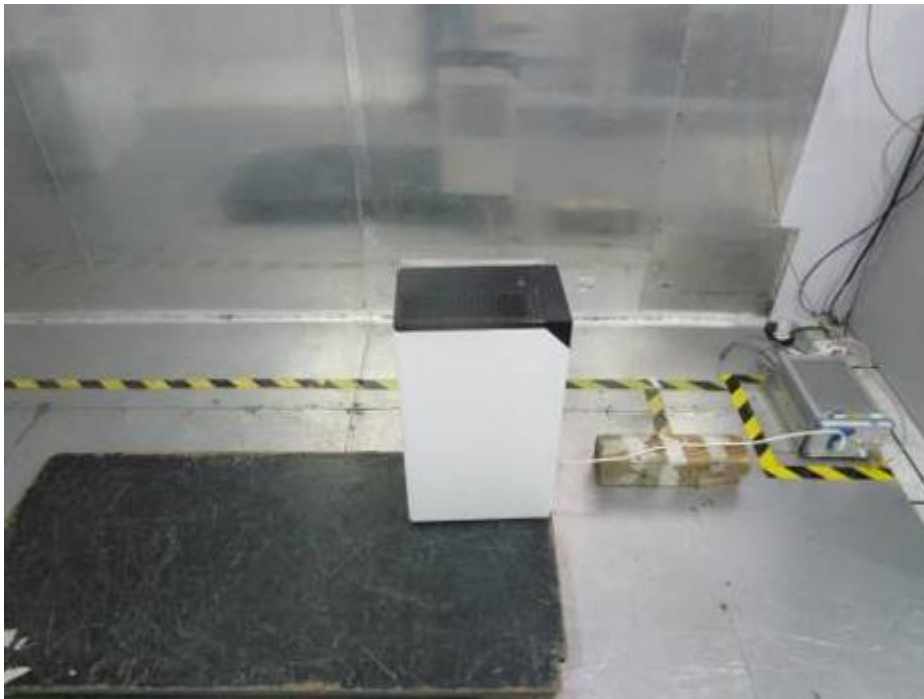
9 Test Setup Photo

Conducted Emissions at AC Mains Power Port (150kHz-30MHz)

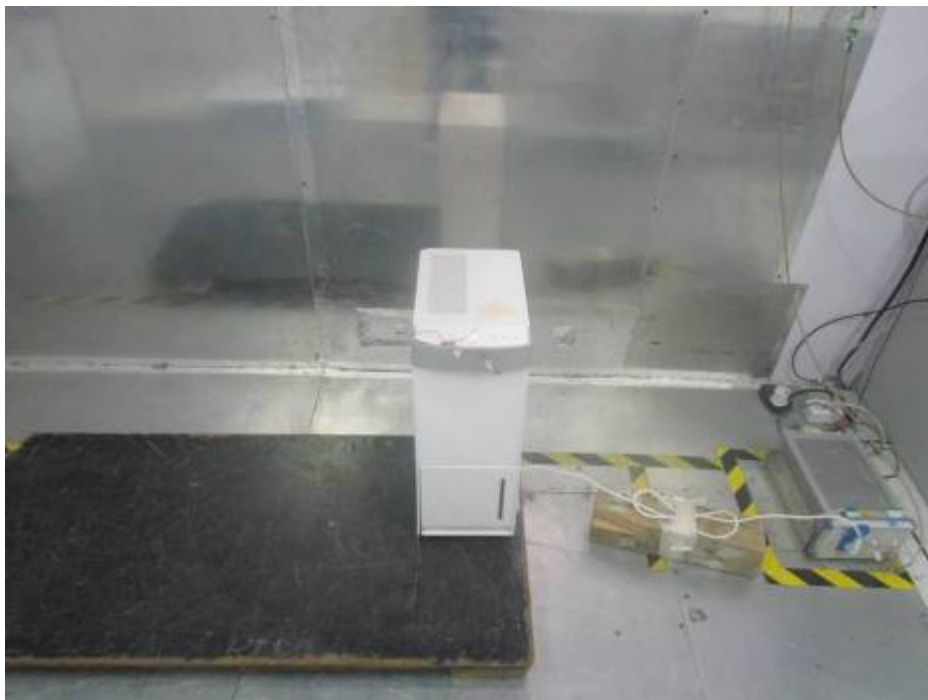
Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



Voltage Fluctuations and Flicker

Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



Radiated Immunity (80MHz-6GHz)

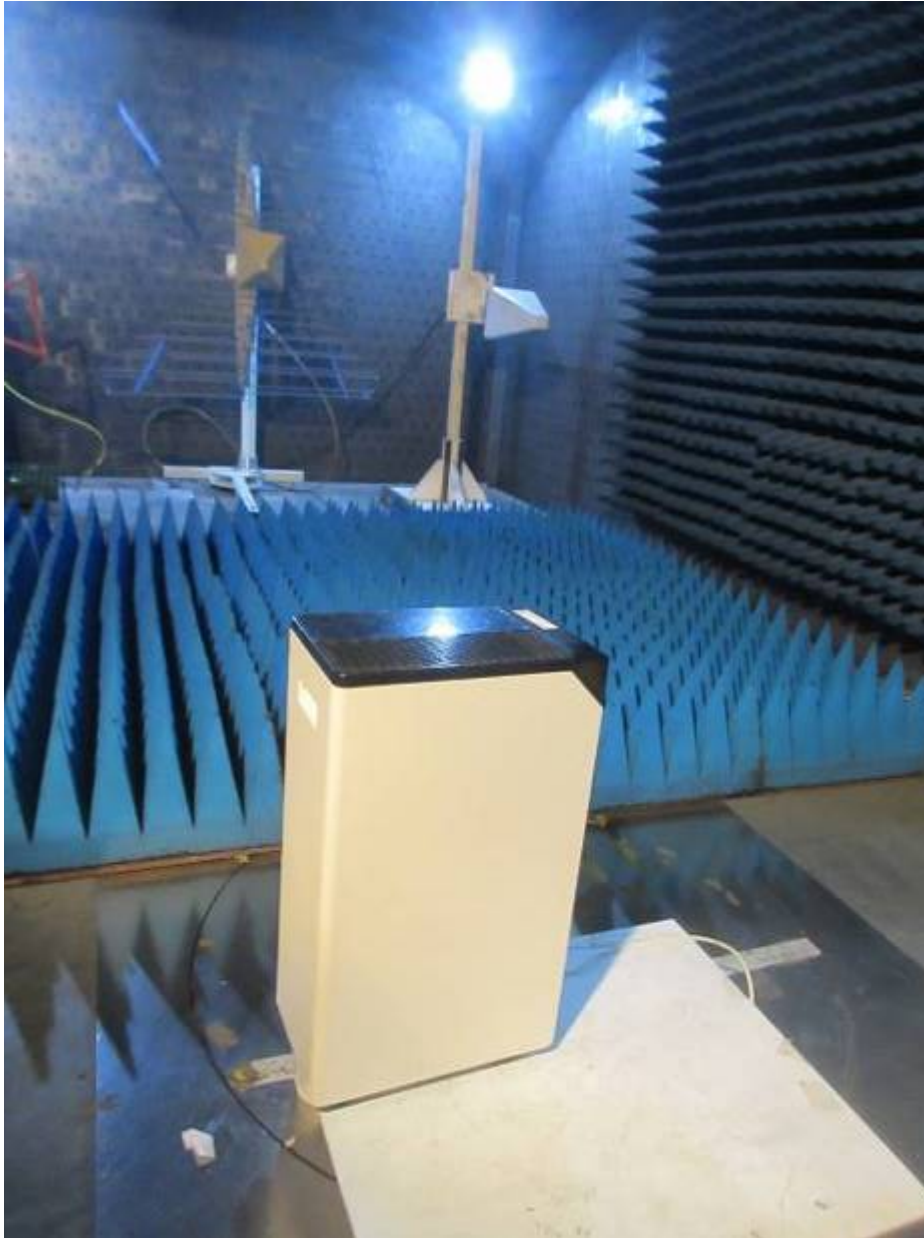
Model: LEO-RCS20(Y)-N7-1



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Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



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Surge at AC Mains Power Port

Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



Conducted Immunity at AC Mains Power Port (150kHz-80MHz)

Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



Voltage Dips and Interruptions

Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



Electrostatic Discharge

Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



Electrical Fast Transients Burst at AC Mains Power Port

Model: LEO-RCS20(Y)-N7-1



Model: LIBRA-RCS20(Y)-N7-1



Model: TAURUS-RCS25(Y)-N7-1



10 EUT Constructional Details (EUT Photos)

Refer to Appendix - Photographs of EUT Constructional Details for GZCR2104020061HS

- End of the Report -